Stewart, Section 15.7, Exercise 22

Problem

Use a triple integral to find the volume of the given solid.

The solid enclosed by the cylinder $x^2 + z^2 = 4$ and the planes $y = -1$ and $y + z = 4$.

Part (a)

Step 1. Name the three boundary surfaces

The surface $x^2 + z^2 = 4$ is a *circular cylinder*. Its generator lines are parallel to the y-axis, and every cross-section perpendicular to that axis is a circle of radius 2 in a plane $y =$ constant. The axis of the cylinder is the y-axis.

The plane $y = -1$ is parallel to the xz-plane. It is the bottom of the solid. The plane $y + z = 4$ is slanted. It is the top of the solid. Write the top plane in the form

$$y = 4 - z.$$

This form is useful because it gives y as a function of z, and z is one of the two variables of the circular cross-section.

The symbol E means the solid, and V means its volume. The symbol dV means the volume element, that is, the volume of a small box with sides dx, dy, and dz in the order that the integration uses.

Step 2. Find the shadow of the solid in the xz-plane

The cylinder is the only surface that restricts x and z. Therefore the shadow of E on the xz-plane is the disk that the cylinder cuts in that plane:

$$D = \{(x, z) : \ x^2 + z^2 \leq 4\}.$$

The disk has radius 2, so its area is

$$\text{area}(D) = \pi \cdot 2^2 = 4\pi.$$

Fix a point (x, z) in D. Draw the line through this point that is parallel to the y-axis. This line hits the plane $y = -1$ first and the plane $y = 4 - z$ last. Both numbers do not depend on y.

So every such line enters E at $y = -1$ and leaves E at $y = 4 - z$, and the length of the part inside E is

$$(4 - z) - (-1) = 5 - z.$$

On the disk we have $-2 \le z \le 2$. Therefore

$$3 \le 5 - z \le 7.$$

The upper limit is always above the lower limit, and the two planes never meet inside the solid. The order $dy\,dz\,dx$ is therefore legal. Figure 1 shows the solid, and Figure 2 shows the disk and the side view.

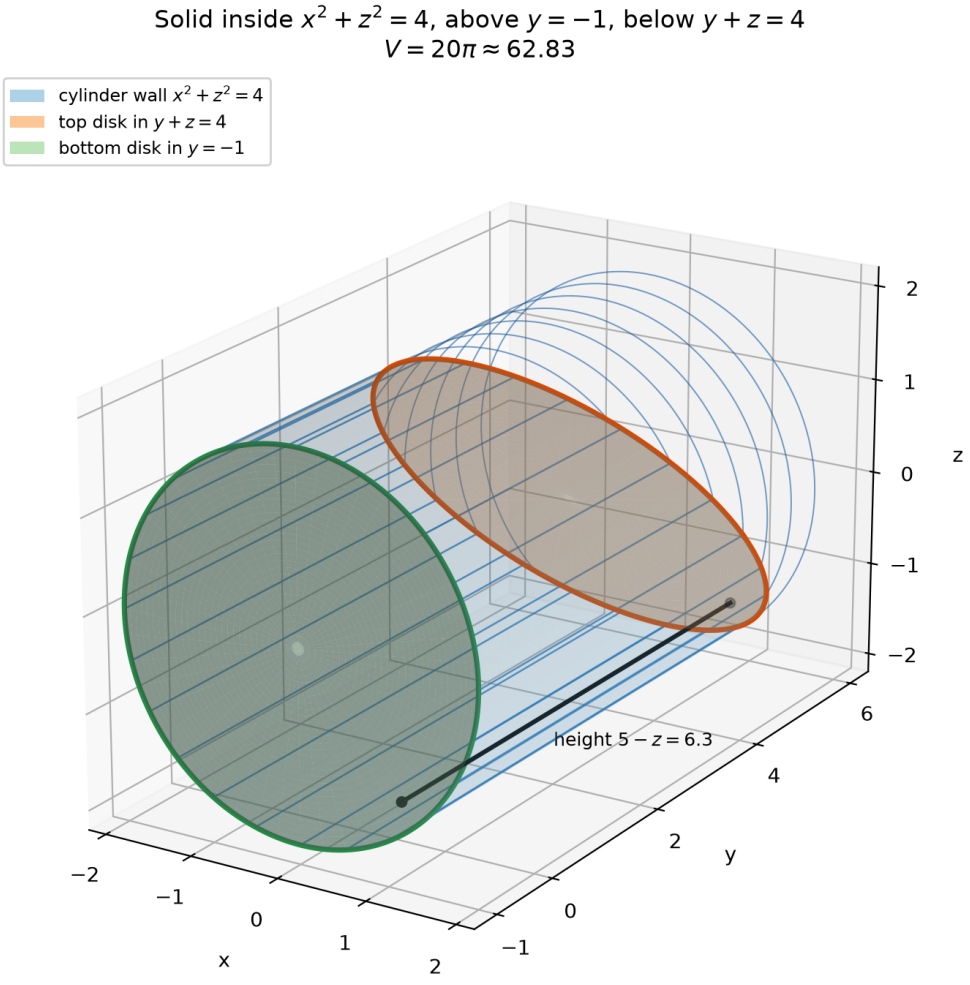


Figure 1: The solid E of Exercise 22. The blue wireframe is the cylinder wall $x^2 + z^2 = 4$. The green disk is the bottom, in the plane $y = -1$. The orange disk is the top, in the slanted plane $y + z = 4$. The top plane cuts the wall in the orange curve, and the bottom is the green circle. The black segment shows the height $5 - z$ at one point of the disk: the wall is longer where z is negative.

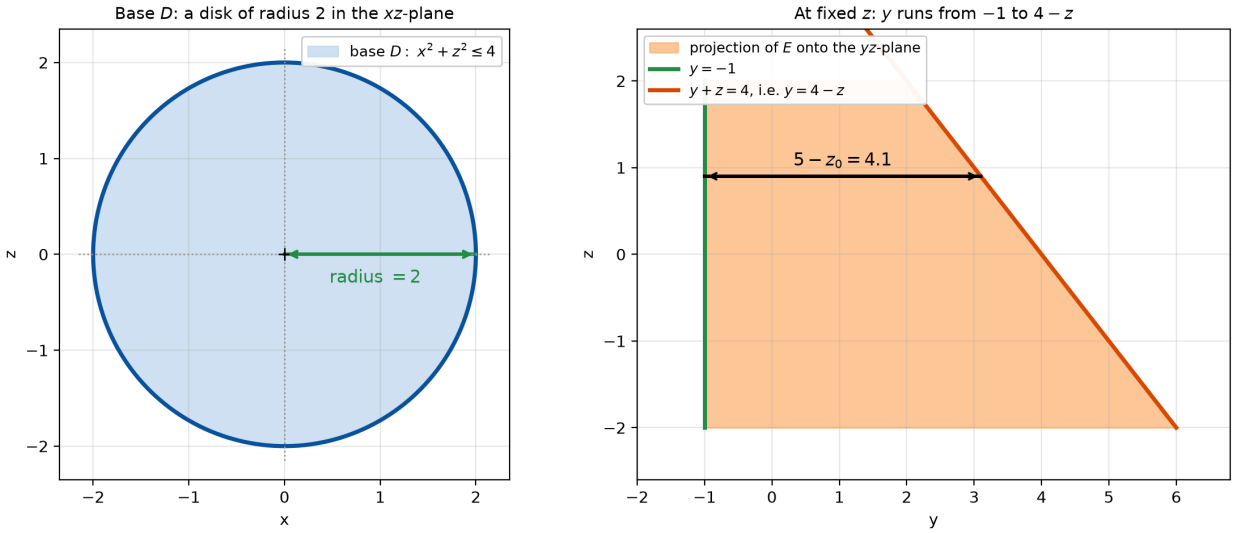


Figure 2: Left: the base disk D in the xz-plane, with radius 2. Right: the projection of E onto the yz-plane. At a fixed z the y-interval is $[-1, 4 - z]$, so the horizontal segment has length $5 - z$. This segment is the same for every x in the disk.

Step 3. Write the triple integral

The inner integral is in y, from -1 to $4 - z$. The outer two integrals run over the disk D, which is described by $-2 \le x \le 2$ and $-\sqrt{4 - x^2} \le z \le \sqrt{4 - x^2}$. So

$$V = \iiint_E 1 \, dV = \int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{-1}^{4-z} dy \, dz \, dx.$$

Step 4. Integrate the inner variable

The inner integral gives the height of the solid above the point (x, z):

$$\int_{-1}^{4-z} dy = \Big[y \Big]_{-1}^{4-z} = 5 - z.$$

So the triple integral is the double integral of the height over the disk:

$$V = \iint_D (5 - z) \, dA.$$

Split the integrand into two terms:

$$V = 5 \iint_D dA - \iint_D z \, dA.$$

The first integral is the area of the disk, 4π. For the second integral, note that the disk is symmetric about the line $z = 0$ and the function z is odd in z. The part above $z = 0$ cancels the part below $z = 0$, so

$$\iint_D z \, dA = 0.$$

Therefore

$$V = 5 \cdot 4\pi - 0 = 20\pi.$$

Step 5. Check the value in cylindrical coordinates

The cylinder invites cylindrical coordinates about the y-axis:

$$x = r\cos\theta, \qquad z = r\sin\theta, \qquad y = y,$$

with $0 \le r \le 2$ and $0 \le \theta \le 2\pi$. The volume element is $dV = r\,dy\,dr\,d\theta$, because the factor r is the length of the arc that the angle $d\theta$ cuts at radius r.

In these variables the plane $y + z = 4$ becomes $y = 4 - r\sin\theta$, so

$$V = \int_0^{2\pi}\int_0^{2}\int_{-1}^{4-r\sin\theta} r\,dy\,dr\,d\theta.$$

The inner integral is

$$\int_{-1}^{4-r\sin\theta} r\,dy = r\big(5 - r\sin\theta\big) = 5r - r^2\sin\theta.$$

Then

$$\int_0^2 \big(5r - r^2\sin\theta\big)\,dr = \left[\frac{5r^2}{2} - \frac{r^3}{3}\sin\theta\right]_0^2 = 10 - \frac{8}{3}\sin\theta.$$

Finally

$$V = \int_0^{2\pi}\left(10 - \frac{8}{3}\sin\theta\right)d\theta = \left[10\theta + \frac{8}{3}\cos\theta\right]_0^{2\pi} = 20\pi.$$

The two routes give the same value, as they must.

Step 6. Check the value with numbers and with bounds

The exact value is

$$20\pi = 62.83185307179586\ldots$$

Three numerical methods in the script `part1_step3.py` confirm this value. Table 1 lists the results.

Method	Value	Error
Midpoint rule in polar coordinates, 2000×2000	62.831853071796	7.1×10^{-15}
Midpoint rule on a 4000×4000 square grid	62.832000000000	1.5×10^{-4}
Monte Carlo, 4×10^6 points	62.759732	7.2×10^{-2}

Table 1: Numerical values of the volume of E. The last error is inside the statistical error $\pm 8.3 \times 10^{-2}$ of the Monte Carlo estimate at three standard deviations.

As a last check, the solid fits inside the box $4 \times 7 \times 4$, which has volume 112. It is also at least 3 units tall over the whole disk, so its volume is at least $3 \cdot 4\pi = 12\pi$. The value 20π lies between 12π and 28π, as it must, and it is about 56% of the box.

Step 7. State the answer

$$\boxed{V = 20\pi \approx 62.83}$$

How to solve problems of this kind

1. **Find the axis of the cylinder.** The variable that is missing from the equation of the cylinder is the direction of its generator lines. Here the equation $x^2 + z^2 = 4$ has no y, so the generator lines are parallel to the y-axis.
2. **Project onto the plane perpendicular to that axis.** The shadow is the disk $x^2 + z^2 \le 4$. This disk is the domain of the two outer integrals.
3. **Read the inner limits from the two planes.** Put the variable of the cylinder axis innermost when the two other surfaces give it explicit bounds. Here the planes give $y = -1$ and $y = 4 - z$.
4. **Check that the limits are in the right order.** The upper limit must be at least the lower limit on the whole shadow. On the disk, $5 - z \ge 3 > 0$, so the order is safe.
5. **Separate the constant part.** Write $5 - z = 5 + (-z)$. The integral of 5 is 5 times the area, and the integral of $-z$ vanishes by symmetry. This step turns a hard-looking integral into two easy ones.
6. **Use the right volume element.** In cylindrical coordinates about the y-axis, $dV = r\,dy\,dr\,d\theta$. The factor r is not optional.
7. **Check the answer three ways.** Use a second coordinate system, a numerical method, and the size of a box that contains the solid.